

Xin (Sean) Zhou

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WORK EXPERIENCE

AI Engineer

2026.05 – present

Independent Contractor

Melbourne, VIC

Ordering and operations platform for To Be Yabby, Melbourne[🔗]. Built the AI agent stack.

- Designed and delivered a **multi-agent AI system** on one shared **Pipecat** voice pipeline, Silero VAD to AssemblyAI **STT** to Claude to Fish Audio **TTS**, serving a customer agent over Twilio and an authenticated staff agent over **WebRTC**, each behind its own isolated tool registry as the security boundary.
- Built the **agentic tool-calling layer** that lets the assistant do the operator's job rather than describe it, a **ReAct loop** over **16 LLM function-calling tools** across orders, stock, menu and daily takings.
- Enforced **human approval** on every order an agent creates, reading back items, total and payment before writing it; withheld irreversible actions from every role including the owner; and hardened both agents against **prompt injection**, with caller identity taken from signed **JWT** claims rather than model input.
- Delivered **multi-modal AI order intake**, turning customer message screenshots into structured orders through **Claude vision** extraction, customer resolution, menu matching and automated form completion.
- Architected a **delegated UI-automation sub-agent** for voice-driven dashboard control, reading the page's **accessibility tree** and driving the matching controls on a smaller model chosen for latency.
- Rebuilt the customer phone line as a **conversational voice AI agent** over a bidirectional Twilio WebSocket, with **barge-in interruption**, tuned end-of-turn detection, and menu-derived keyterms for bilingual speech, retiring a request-response design the caller could not interrupt.
- Delivered **multi-turn session continuity** across mid-sentence network drops, rejoining the same **LiveKit room** with transcript and tool context intact, and every session persisted as one **auditable tool-call trace**.
- Filed an upstream **issue and pull request** on **Pipecat's** LiveKit transport[🔗], with a reproduction harness; the framework's own fix cites it, and adopted a second defect from the review[🔗].
- Shipped and ran the **agent stack** in production, deployed by **GitHub Actions** onto **AWS Lightsail**, with a **self-hosted staging environment** on two deploy slots so a change was exercised against a live agent first.
- Kept the **agent service** releasable while calls were live, with a **drain on shutdown** so a deploy never strands a conversation, and eliminated a **4.5-second cold start** by preloading models and the inference service.

Software Engineer · Intern

2024.11 – 2025.10

Museums Victoria

Melbourne, VIC

Great Melbourne Telescope restoration at Scienceworks Museum. Built the AI document retrieval system.

- Architected a **hybrid RAG system** over an engineering design archive, combining semantic chunking, BM25 keyword and dense embedding retrieval, and rank fusion, achieving **95% top-5 accuracy at 8 ms query latency**.
- Built a **grounded question-answering service** with per-sentence citation, abstention on out-of-scope queries, and **MCP integration** exposing three retrieval tools to coding agents, served through a FastAPI console.
- Engineered a **LangGraph verification agent** adjudicating documentation claims against source code as ground truth, with **conditional-edge retry**, disconfirming-evidence search, and a bounded two-round search loop.

PROJECT EXPERIENCE

StyleAI[🔗]

2025.03 – 2025.07

Personal styling system built by a four-person team. Co-designed the pipeline and owned deployment.

- Co-designed a **multi-modal styling pipeline**, a **vision LLM** to **DistilBERT** attribute embeddings to a Fashn.ai **generative virtual try-on**, turning a single user photo into a rendered outfit.
- Architected the retrieval stage as **attribute-level matching** over **sentence embeddings**, scoring each attribute independently rather than one vector per image, then a **re-ranking pass** to a single result.
- Deployed the system self-hosted, **Docker** services behind **nginx** and an outbound-only **Cloudflare Tunnel**, running inference as polled background jobs so slow model calls stay off the request path.

Personal site, AI agent service, Cloudflare Worker and job tracker, built and operated end to end across AWS, Google Cloud and self-managed Linux hosts. Sole author and operator.

- Built **continuous deployment** for the entire platform on **GitHub Actions** with **self-hosted runners**, releasing to **GitHub Pages**, **Cloudflare** and the Linux hosts that run the **AI agent service**.
- Integrated **Amazon Bedrock** as an alternative inference provider for the production **AI agent**, behind a single **tool-calling** interface, removing dependence on any one model vendor.
- Ran a scheduled **adversarial test suite** against the deployed agent for **prompt injection** and jailbreak behaviour, split out of the main pipeline because behavioural cases need a real model call.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript
AI & Agentic Systems	Multi-agent architecture, RAG / hybrid retrieval, MCP, LangGraph, Multi-modal vision
Real-Time Voice	WebRTC, WebSocket, Streaming STT / TTS, LiveKit
Web Development	FastAPI, PostgreSQL, Next.js, React, Server-Sent Events
Infrastructure	Docker, GitHub Actions, self-hosted Linux
Practices	Domain-Driven Design, Role-based access control

EDUCATION

University of Melbourne – Master of Software Engineering	Grade H2A	2023.07 – 2026.06
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